

Project Euler Problem 1: Multiples of 3 or 5

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Problem

If we list all the natural numbers below 10 that are multiples of 3 or 5, we get 3, 5, 6 and 9. The sum of these multiples is 23.

Find the sum of all the multiples of 3 or 5 below 1000.

Proof

Rather than testing every number below n , we sum the multiples of 3 and of 5 directly and correct for the numbers counted twice. Throughout, we use the closed form for the sum of the first k integers, proven by induction in Problem 6:

$$T(k) = \sum_{j=1}^k j = \frac{k(k+1)}{2}.$$

Definition 1. For $d, n \in \mathbb{N}$, let $S_d(n)$ be the sum of the positive multiples of d below n .

Lemma 2. For all $d, n \in \mathbb{N}$,

$$S_d(n) = d \cdot T\left(\left\lfloor \frac{n-1}{d} \right\rfloor\right).$$

Proof. The positive multiples of d are jd for $j \geq 1$. Such a multiple is below n exactly when $jd \leq n-1$, that is, when $j \leq \frac{n-1}{d}$. Since j is an integer, this is equivalent to $j \leq k$, where $k = \left\lfloor \frac{n-1}{d} \right\rfloor$. Hence the multiples of d below n are $d, 2d, \dots, kd$, and

$$S_d(n) = \sum_{j=1}^k jd = d \sum_{j=1}^k j = d \cdot T(k).$$

If $k = 0$, there are no such multiples, and both sides are 0. $\square$

Definition 3. For a finite nonempty set D of positive integers, the *least common multiple* $\text{lcm}(D)$ is the smallest positive integer that is a multiple of every element of D .

For this definition to make sense, a smallest such integer must exist. Let $P = \prod_{e \in D} e$ be the product of all elements of D . For every $d \in D$,

$$P = d \cdot \prod_{e \in D \setminus \{d\}} e,$$

where the second factor is a product of positive integers and is therefore a positive integer (it equals 1 when $D = \{d\}$, since an empty product is 1). Hence $d \mid P$ for every $d \in D$, and $P > 0$ as a product of positive integers. So the set of positive common multiples of D ,

$$M = \{m \in \mathbb{Z}_{>0} : d \mid m \text{ for every } d \in D\},$$

contains P and is nonempty. By the well-ordering principle, M has a least element, and that element is $\text{lcm}(D)$. Hence $\text{lcm}(D)$ is well defined.

Lemma 4. *A positive integer x is a multiple of all elements of a finite nonempty set D of positive integers if and only if it is also a multiple of $\text{lcm}(D)$.*

Proof. Let $m = \text{lcm}(D)$. We first show that if x is a multiple of m , then it is a multiple of every element of D . Say $x = km$ for some integer k . Let $d \in D$. By definition 3, $d \mid m$, so there is an integer p with $m = pd$. Then

$$x = km = k(pd) = (kp)d,$$

and kp is an integer, so $d \mid x$. Since d was arbitrary, x is a multiple of every element of D .

Conversely, we show that if x is a multiple of every element of D , then it is necessarily a multiple of m as well. Using the set M defined right after Definition 3, we know that m is the smallest multiple of every element of D and that m is greater than 0. We start by dividing x by m :

$$\begin{aligned} x &= qm + r, & 0 \leq r < m, \\ r &= x - qm. \end{aligned}$$

For an arbitrary d in D , we know that $d \mid x$ by the hypothesis. Since $m \in M$, we also know that $d \mid m$, implying that $d \mid qm$. So there are integers a and b with $x = ad$ and $qm = bd$, and

$$r = x - qm = ad - bd = (a - b)d.$$

Therefore, $d \mid r$, and since d was arbitrary, every element of D divides r . If we suppose that $r > 0$, then r would be a member of M that is smaller than m . But m is defined to be the smallest member of M , so $r = 0$. This means that $x = qm$, where q is an integer, making x a multiple of m . $\square$

With only two divisors, it is easy to see that subtracting the multiples of 15 once fixes the double counting. The same idea works for any set of divisors, but with more of them the overlaps are harder to track by hand, so we prove the general statement by looking at how many times each number is counted.

Lemma 5. *Let D be a finite nonempty set of positive integers and let $n \in \mathbb{N}$. The sum of the positive integers below n that are multiples of at least one element of D is*

$$\sum_{\emptyset \neq A \subseteq D} (-1)^{|A|+1} S_{\text{lcm}(A)}(n).$$

Proof. The sum we want adds each x below n that is a multiple of some element of D exactly once, and leaves out every other x . We rewrite the alternating sum in the statement as a single sum over x , where each x is multiplied by a coefficient that we call its *weight*. We then show that the weight of x is 1 if x is a multiple of some element of D and 0 otherwise, so the two sums agree.

For each integer x with $1 \leq x < n$, let $D_x = \{d \in D : d \mid x\}$ be the set of divisors in D that divide x , and let $m_x = |D_x|$.

Let $A \subseteq D$ be nonempty. Then A is a finite nonempty set of positive integers, so by lemma 4, a positive integer is a multiple of every element of A if and only if it is a multiple of $\text{lcm}(A)$. So $S_{\text{lcm}(A)}(n)$ is the sum of those $x < n$ that are divisible by every element of A , that is, those with $A \subseteq D_x$. Writing $[P]$ for 1 if the statement P is true and 0 otherwise, we can sum over every candidate below n and keep only those that qualify:

$$S_{\text{lcm}(A)}(n) = \sum_{x=1}^{n-1} x [A \subseteq D_x].$$

Substituting this, swapping the order of the two finite sums, and factoring out x , which does not depend on A ,

$$\begin{aligned} \sum_{\emptyset \neq A \subseteq D} (-1)^{|A|+1} S_{\text{lcm}(A)}(n) &= \sum_{\emptyset \neq A \subseteq D} (-1)^{|A|+1} \sum_{x=1}^{n-1} x [A \subseteq D_x] \\ &= \sum_{x=1}^{n-1} x \sum_{\emptyset \neq A \subseteq D} (-1)^{|A|+1} [A \subseteq D_x] \\ &= \sum_{x=1}^{n-1} x \sum_{\emptyset \neq A \subseteq D_x} (-1)^{|A|+1}. \end{aligned}$$

The last step uses the bracket to drop the subsets that are not contained in D_x . The inner sum is the weight of x . To compute it, we group the nonempty subsets of D_x by size. For each j with $1 \leq j \leq m_x$, there are $\binom{m_x}{j}$ subsets of size j , and each contributes $(-1)^{j+1}$. Hence

$$\sum_{\emptyset \neq A \subseteq D_x} (-1)^{|A|+1} = \sum_{j=1}^{m_x} (-1)^{j+1} \binom{m_x}{j} = w(m_x),$$

where for any integer $k \geq 0$ we define

$$w(k) = \sum_{j=1}^k (-1)^{j+1} \binom{k}{j}.$$

If $k = 0$, the sum is empty and $w(0) = 0$. If $k \geq 1$, then

$$\begin{aligned}
0 &= (1 - 1)^k = \sum_{j=0}^k \binom{k}{j} 1^{k-j} (-1)^j && \text{by the binomial theorem} \\
&= \sum_{j=0}^k \binom{k}{j} (-1)^j && \text{since } 1^{k-j} = 1 \\
&= 1 + \sum_{j=1}^k \binom{k}{j} (-1)^j && \text{pulling out the } j = 0 \text{ term} \\
&= 1 - \sum_{j=1}^k \binom{k}{j} (-1)^{j+1} && \text{since } (-1)^j = -(-1)^{j+1} \\
&= 1 - w(k),
\end{aligned}$$

so $w(k) = 1$. Thus $w(m_x) = 1$ when $m_x \geq 1$, that is, when x is a multiple of some element of D , and $w(m_x) = 0$ otherwise. Therefore

$$\sum_{\emptyset \neq A \subseteq D} (-1)^{|A|+1} S_{\text{lcm}(A)}(n) = \sum_{x=1}^{n-1} x w(m_x) = \sum_{\substack{1 \leq x < n \\ m_x \geq 1}} x,$$

which is the sum of the positive integers below n that are multiples of at least one element of D . $\square$

Theorem 6. *For all $n \in \mathbb{N}$, the sum of the positive integers below n that are multiples of 3 or 5 is*

$$S_3(n) + S_5(n) - S_{15}(n).$$

Proof. Apply lemma 5 with $D = \{3, 5\}$. The nonempty subsets are $\{3\}$ and $\{5\}$, each of size 1, and $\{3, 5\}$, of size 2. A single-element set $\{d\}$ has $\text{lcm}(\{d\}) = d$, since d is the smallest positive multiple of d . For $\{3, 5\}$, the number 15 is a positive multiple of both 3 and 5. Any smaller common multiple would be a positive multiple of 5 below 15, so it would be 5 or 10, and 3 divides neither. Hence $\text{lcm}(\{3, 5\}) = 15$, and the three terms are $S_3(n)$ and $S_5(n)$ with sign $+$, and $S_{15}(n)$ with sign $-$. $\square$

Answer

As a check, the example from the problem statement ($n = 10$) gives

$$3 \cdot T(3) + 5 \cdot T(1) - 15 \cdot T(0) = 18 + 5 - 0 = 23.$$

For $n = 1000$, we have $\lfloor 999/3 \rfloor = 333$, $\lfloor 999/5 \rfloor = 199$, and $\lfloor 999/15 \rfloor = 66$. By theorem 6 and lemma 2,

$$\begin{aligned}
 S_3(1000) + S_5(1000) - S_{15}(1000) &= 3 \cdot T(333) + 5 \cdot T(199) - 15 \cdot T(66) \\
 &= 3 \cdot 55611 + 5 \cdot 19900 - 15 \cdot 2211 \\
 &= 166833 + 99500 - 33165 \\
 &= 233168.
 \end{aligned}$$